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Bicycle hydraulic disc braking system and a method and system for diagnosing the braking system

Abstract

A diagnostic method for a bicycle hydraulic disc braking system, having a master cylinder assembly with a cylinder inside which a piston reciprocates against the action of a return spring. The master cylinder has a hydraulic tank with a bottom, a ceiling, and a membrane that divides the hydraulic tank into two chambers. The hydraulic tank is connected to the cylinder through the bottom. A metering device measures the distance between the membrane and the ceiling of the tank. The output signal of the metering device is evaluated to determine that operation is correct when the membrane temporarily approaches the ceiling by a predetermined extent range, or incorrect otherwise. A positive evaluation reflects the correct operating condition, and a negative evaluation reflects an incorrect operating condition.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application claims the benefit of Italian Patent Application No. 102021000020564, filed on Jul. 30, 2021, which is incorporated herein by reference as if fully set forth.
FIELD OF INVENTION
(2) The invention relates in general to the field of bicycle hydraulic braking systems of the disc type.
BACKGROUND
(3) The invention relates in particular to a diagnosis system and method for a bicycle hydraulic braking system of the disc type, in particular for a master cylinder assembly thereof, as well as to a manual control device of such a system, the braking system, and a bicycle equipment comprising the system.
(4) A bicycle braking system of the disc type generally comprises a braking device, also called brake caliper, at either or each wheel, wherein friction elements, also called pads, at least one of which is movable, are hydraulically brought into engagement with a disc integrally rotating with the hub of the bicycle wheel, in order to brake it by friction.
(5) Bicycle disc braking systems can be of a hydraulic type.
(6) These systems also require a hydraulic fluid, also called brake fluid, contained in a sealed circuit of the system.
(7) In hydraulic braking systems, the hydraulic fluid circuit comprises a circuit active part and a tank in fluid communication with the circuit active part, to compensate for changes in the amount of hydraulic fluid in the active part of the circuit. Said amount changes may be due to several factors, including the operating temperature, the displacement of moving members, and the wear of the friction elements.
(8) Typically, the tank is part of a master cylinder assembly further comprising a cylinder and a piston movable by reciprocating motion inside the cylinder against the action of a return spring, wherein the tank is fluidically connected to the cylinder through at least one main passage, which is closed or open according to the position of a head of the piston, typically provided with a gasket.
(9) EP3835191A1 discloses a hydraulic tank for a bicycle comprising: a first wall; a second wall opposite to the first wall and configured to place the tank in liquid communication with a hydraulic cylinder; a side wall that extends between the first wall and the second wall; an elastically

deformable membrane operating between the first wall and the second wall and defining inside the tank a compensation chamber having a volume variable between a maximum volume defined in a condition of maximum expansion of the compensation chamber and a minimum volume defined in a condition of minimum expansion of the compensation chamber; wherein the membrane comprises a perimeter fixing portion which is fixed to the tank, a thrusting portion configured to act on the brake liquid provided in the tank and a perimeter inlet portion inside the tank arranged between the perimeter fixing portion and the thrusting portion, wherein in the condition of minimum expansion of the compensation chamber the thrusting portion is completely arranged between the perimeter inlet portion of the membrane and the second wall of the tank.

(10) U.S. Pat. No. 10,082,158B2 discloses a master mounting for a hydraulic actuation member which can be used for hydraulic brakes or clutch actuations, comprising, inter alia: a piston; a housing defining a compensating chamber having an interior; a cylinder having a cylinder wall, the piston being guided in the cylinder, a communication channel having an opening in the cylinder wall and connecting the cylinder and the compensating chamber in at least one position of the piston, and at least one overflow channel disposed at least at the opening of the communication channel in the cylinder wall; a gasket having an outer surface and being disposed between the piston and the cylinder; a cover; and a bellows within the compensating chamber and closed therein by the cover.

(11) The Applicant has recognized that an incorrect mounting of the piston or of its gasket or a severe wear or breaking of the latter, as well as a clogging of said main passage due to any reason, may cause the total malfunction of the braking system, with consequent severe risk of falling of the cyclist and of road accidents.

(12) The technical problem at the basis of the invention is to allow the diagnosis of the hydraulic system, in particular of a master cylinder assembly thereof.

SUMMARY

(13) A diagnostic method for a bicycle hydraulic braking system, and in particular a disc brake system, having a master cylinder assembly with a piston that reciprocates against the action of a return spring. The master cylinder has a hydraulic tank with a bottom, a ceiling, and a membrane that divides the hydraulic tank into two chambers. The hydraulic tank is connected to the cylinder through the bottom. A device measures the distance between the membrane and the ceiling of the tank, and outputs a signal based on the measurement. The output signal is evaluated in a data processor to determine that operation is correct when the membrane temporarily approaches the ceiling by a predetermined extent range, or incorrect otherwise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further features and advantages of the invention will be better highlighted by the description of example embodiments thereof, made with reference to the attached drawings, wherein:

(2) FIG. **1** schematically shows a bicycle hydraulic braking system,

(3) FIGS. **2-15** schematically show some components of the bicycle hydraulic braking system, in different operating conditions,

(4) FIG. **16** schematically shows a hydraulic tank in different operating conditions, and associated sensors,

(5) FIG. **17** is analogous to FIG. **16**, but wherein the hydraulic tank has sensors of a different type,

(6) FIGS. **18** and **19** show a membrane of the hydraulic tank, respectively in a perspective view and in a partially sectional perspective view,

(7) FIGS. **20** and **21** schematically show different faces of a printed circuit board which may be used in the bicycle hydraulic system,

(8) FIG. **22** schematically shows a different printed circuit board which may be used in the bicycle hydraulic system,
(9) FIG. **23** shows a control device part of a bicycle hydraulic braking system,
(10) FIG. **24** schematically shows a section across the control device of FIG. **23**,
(11) FIG. **25** shows a master cylinder assembly of a bicycle hydraulic system and a kinematic associated thereto, and
(12) FIGS. **26-27** show a braking device of a bicycle hydraulic system, respectively without a friction element and provided therewith, wherein FIG. **27** is cutaway.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) In one aspect, the invention relates to a diagnosis method for a bicycle hydraulic braking system of the disc type, in particular of a master cylinder assembly thereof, the method comprising the steps of: a) providing a master cylinder assembly comprising a cylinder, a piston movable by reciprocating motion inside the cylinder against the action of a return spring, as well as a hydraulic tank comprising a bottom, a ceiling, and a membrane dividing the tank in a variable volume filled chamber and a complementarily variable volume empty chamber, said tank being fluidically connected to the cylinder through at least one main passage in said bottom, b) providing a meter device of the distance of the membrane from the ceiling of the tank, c) inserting a hydraulic fluid in the master cylinder assembly, d) moving the piston by reciprocating motion inside the cylinder, e) evaluating, with processing means, whether an output signal of the membrane detector device is indicative of a temporary approach of the membrane to the ceiling having an extent corresponding to a predetermined approach range, f) establishing that the master cylinder assembly is correctly operating if the evaluation is positive, that the master cylinder assembly is not correctly operative if the evaluation is negative.

(14) In the present description and in the attached claims, under term “diagnosis” the examination aimed at formulating a judgment on the conditions and the operation of the various parts is meant to be indicated, including testing.

(15) The Applicant observed that if the master cylinder assembly is correctly operating, in particular if said main passage is not clogged and there is sealing between the piston and the cylinder, in particular if a gasket possibly arranged at a head of the piston is undamaged, the thrust of the piston first forces a part of the brake fluid in the tank and only thereafter, if and when said main passage is correctly closed at crossing thereof by the piston head, possibly provided with a gasket, the brake fluid is forced in the active part of its circuit. On the other hand, if the passage is clogged the brake fluid is immediately forced in the circuit active part, without being initially forced in the tank; if there is no suitable sealing, the brake fluid is not (or not only) forced in the circuit active part and is instead (or also) forced in a larger amount in the tank.

(16) Through the evaluation of the distance of the membrane from the ceiling, it is therefore possible to control that said behavior takes place correctly.

(17) The master cylinder assembly may be fluidically connected in the hydraulic braking system during the execution of steps d), e), f).

(18) In an aspect, the invention relates to a diagnosis system of a bicycle hydraulic braking system of the disc type or of a master cylinder assembly thereof, the system comprising: a master cylinder assembly comprising a cylinder, a piston movable by reciprocating motion inside the cylinder against the action of a return spring, as well as a hydraulic tank comprising a bottom, a ceiling, and a membrane dividing the hydraulic tank in a variable volume filled chamber and a complementarily variable volume empty chamber, said tank being fluidically connected to the cylinder through at least one main passage in said bottom, a hydraulic fluid filing at least in part the master cylinder assembly, a meter device of the distance of the membrane from the ceiling of the tank, a data processing system configured to evaluate whether an output signal of the meter device is indicative of a temporary approach of the membrane to the ceiling having an extent corresponding to a predetermined approach range, and to establish that the master cylinder assembly is correctly

operating if the evaluation is positive, that the master cylinder assembly is not correctly operating if the evaluation is negative.
(19) The tank may comprise a lateral wall extending between the ceiling and the bottom.
(20) The temporary approach of the membrane to the ceiling may occur during at least one length of the spring compression stroke of the piston and/or of the return stroke.
(21) The temporary approach may start and end at the crossing of the main passage by the head of the piston in the cylinder of the master cylinder assembly.
(22) The distance meter device may comprise a sensor arranged at the ceiling of the tank and possibly a detectable member borne by the membrane.
(23) The sensor may be an optical sensor, comprising a light source, preferably an LED, and a photodiode or phototransistor located to receive the light emitted by the light source and reflected by the membrane or by said detectable member borne by the membrane when it is in the field of view of the optical sensor.
(24) The detectable member borne by the membrane may be an insert of more reflective material than the material forming the membrane.
(25) The optical sensor may be located on the inner surface of the ceiling of the tank, or it may be located on its outer surface in case the ceiling is transparent.
(26) In the membrane a pocket may be formed configured to house the detectable member borne by the membrane.
(27) The pocket may comprise a region wherein two layers of the membrane overlap each other and a cut in one of said overlapping layers.
(28) The membrane may have a peripheral region intended to be fixed to the tank and a deformable central region, which in general adopts a curved configuration, with variable radius of curvature.
(29) The sensor may face a region of the membrane as central as possible, namely as far as possible from its peripheral region fixed to the tank.
(30) The meter device may be selectively actuatable by the cyclist.
(31) Alternatively or additionally, the membrane detector device may be only periodically actuated under the control of said data processing system.
(32) In this case, the diagnosis system may have a stand-by mode and comprise a waking system.
(33) The diagnosis system may comprise at least one output device of the outcome of said evaluation.
(34) Each of said at least one output device may be selected from the group consisting of: at least one audible indicator, at least one luminous indicator, and a communication device.
(35) Said at least one sensor and said at least one luminous indicator, if present, may be housed on opposite faces of one and a same printed circuit board.
(36) The master cylinder assembly may comprise a gasket arranged at a piston head.
(37) In an aspect, the invention relates to a control device of a bicycle hydraulic braking system of the disc type, comprising a manual actuation member configured to issue a braking command and the diagnosis system described above, wherein the piston is displaced inside the cylinder in a forward stroke in response to the actuation of the manual actuation member.
(38) The manual actuation member is thus configured to apply, directly or indirectly, a force when subject to a predetermined manual action, and the master cylinder assembly is configured to convert the force into pressure of the hydraulic fluid in said hydraulic braking system.
(39) In an aspect, the invention relates to a bicycle hydraulic braking system of the disc type, comprising: at least one control device, comprising a manual actuation member configured to issue a braking command, the master cylinder assembly and the meter device of the diagnosis system described above, wherein the piston is displaced inside the cylinder in a forward stroke in response to the actuation of the manual actuation member, and a respective hydraulic braking device fluidically connected to the master cylinder assembly, wherein the data processing system of said diagnosis system is part of a component of the hydraulic braking system, preferably is part of the

control device.
(40) In the hydraulic braking system, the control device and the braking device are fluidically connected through a conduit to form a brake fluid circuit and, in a condition of use, a brake fluid is sealed and under vacuum in the brake fluid circuit.
(41) The braking device may comprise at least one slave cylinder assembly.
(42) The slave cylinder assembly may comprise a cylinder, a piston movable by reciprocating motion inside the cylinder and a friction element moved by the piston.
(43) In another aspect, the invention relates to a bicycle equipment comprising: a bicycle hydraulic braking system of the disc type, comprising: at least one control device, comprising a manual actuation member configured to issue a braking command, the master cylinder assembly and the meter device of the diagnosis system described above, wherein the piston is displaced inside the cylinder in a forward stroke in response to the actuation of the manual actuation member, and a respective hydraulic braking device fluidically connected to the master cylinder assembly, and a speed gear shifting system, said at least one control device comprising at least one second manual actuation member configured to issue a speed gear shifting command, wherein the data processing system of said diagnosis system is part of a component of the speed gear shifting system, for example it is part of gearshift assembly thereof associated with the hub of the rear wheel.
(44) As far as the hydraulic braking system is concerned, the above considerations apply.
(45) Said data processing system may comprise a micro-controller.
(46) Alternatively or additionally, said data processing system may comprise electric components and/or discrete electronic components.
(47) FIG. **1** shows, in a totally schematic manner, a bicycle hydraulic braking system **1** of the disc type according to an embodiment of the invention.
(48) The braking system **1** shown comprises a manual control device **2** and a braking device **3** of a wheel, also called brake caliper, fluidically connected through a conduit **4** to form brake fluid circuit, as well as, in a condition of use, a hydraulic fluid **5**, also called brake fluid **5**, sealed and under vacuum within the brake fluid circuit. In the figures, wherever possible the brake fluid **5** is schematized by an oblique fill, in either direction.
(49) Those skilled in the art will understand that typically the bicycle braking system comprises a pair of braking devices **3**, respectively associated with the front wheel and the rear wheel of the bicycle, as well as a pair of control devices **2**, one associated with the braking device **3** of the front wheel and typically mounted at the left grip of the handlebars, and the other one associated with the braking device **3** of the rear wheel and typically mounted at the right grip of the handlebars—although other mounting positions of the control devices **2** are generally possible.
(50) The control device **2** comprises a manual actuation member **6** configured to apply, directly or indirectly, a force when subject to a predetermined manual action, and an actuator or master cylinder assembly **7**, for converting the force into pressure of the brake fluid **5**.
(51) The master cylinder assembly **7** comprises a cylinder **8** and a piston **9** movable by reciprocating motion inside the cylinder **8**, against the action of a return spring **10**. Gaskets **11**, **12** may be arranged at the head **13** and at the thrust end **14** of the piston **9**, respectively. The gaskets **11**, **12**, only schematically shown in FIG. **1**, are preferably of the so-called “lip seal ring” type.
(52) The master cylinder assembly **7** further comprises a tank **20** fluidically connected to the cylinder **8**, configured to contain a supply of brake fluid **5**. The tank **20** and the cylinder **8** are typically fluidically connected through a main passage **21** and a lubrication passage **21***a*.
(53) The tank comprises a “ceiling” **22** and a bottom **23**.
(54) In the description and the attached claims, under “ceiling” of the tank it is meant to indicate the wall of the tank opposed to the wall of the tank comprising the main passage, herein indicated as “bottom” of the tank. Those terms are thus defined solely with reference to the tank itself, and they should not be understood as absolute spatial references because the bottom of the tank and the ceiling of the tank do not necessarily extend in horizontal planes. Indeed, in practice they have

different inclinations.
(55) The tank **20** is represented as if it were a straight rectangular prism, further comprising a lateral wall **24**, but in practice it may have a different shape.
(56) In the present description and in the attached claims, under the expression "lateral wall" of the tank, all the region of the wall of the tank **20** extended between the ceiling **22** and the bottom **23** should be understood.
(57) The manual actuation member **6** is typically of the lever type, as shown in a totally schematic manner. A suitable kinematics (not shown in FIG. **1**, but cf. for example the kinematics **131** of FIG. **25**) may be interposed between the manual actuation member **6** and the master cylinder assembly **7**, in particular between the manual actuation member **6** and the piston **9**.
(58) The pull of the lever or in general the manual action on the manual actuation member **6** determines the thrust of the piston **9** of the master cylinder assembly **7** in a direction compressing the brake fluid **5** and the return spring **10**. When the pull on the lever is released, or in general when the manual action on the manual actuation member **6** ceases, the return spring **10** decompresses, determining the thrust of the piston **9** in the opposed direction.
(59) The tank **20** allows the amount of brake fluid **5** contained downstream of the head **13** of the piston **9** to change, as better discussed hereinbelow. Because the circuit of the brake fluid **5** has to remain under vacuum, a membrane **25** is provided in the tank **20**, dividing the tank **20** in a variable volume filled chamber **26** and a complementarily variable volume empty chamber **27**. The membrane **25** remains in contact with the free surface of the brake fluid **5**.
(60) A vent hole **28** is made in the tank **20** above the membrane **25**, for example in the ceiling **22** thereof, in order to ensure that the pressure on the membrane **25** from the side of the empty chamber **27** is the atmospheric one.
(61) In FIG. **1**, as well as in FIGS. **2-15** described hereinafter, the membrane **25** is schematically represented by a straight line in different positions, but in practice the membrane **25** may be for example fixed in position within the tank **20** at a peripheral region thereof, and have a deformable central region, which in general adopts a curved configuration, having a variable radius of curvature. See, again by way of an example only, FIG. **18-19** subsequently described.
(62) The braking device **3** comprises one or more actuators or slave cylinder assemblies **30**, fluidically connected to conduit **4**, and a respective friction element **31**, or pad **31**, mobile, through the respective slave cylinder assembly **30**, into engagement and out of engagement with a disc W integrally rotating with the hub (not shown) of the bicycle wheel to which the braking device **3** is associated.
(63) The braking device **3** shown merely by way of an example in FIG. **1** comprises a pair of slave cylinders **30** supported in a fixed mutual position, and having substantially parallel and converging compression directions, and a pair of pads **31** which can be shifted from a rest position wherein they are at a certain distance from each other, forming a space or gap **35** wherein the disc W is free to rotate, and a close position wherein they are in contact with and thrusting on the disc W in order to brake it by friction. However, on either side of the disc W, the friction element **31** or pad might be fixed.
(64) The or each slave cylinder assembly comprises a cylinder **32** and a piston **33** movable by reciprocating motion inside the cylinder **32** for a stroke, variable as better discussed hereinbelow. A gasket **34**, typically with square cross-section ("quad ring" or Q-ring"), is interposed between the piston **33** and the cylinder **32**.
(65) When the braking system **1** is undamaged, the amount of brake fluid **5** is correct and rather it is, as an assumption, at the maximum allowed, and the pads **31** are undamaged, the operation of the braking system **1**, neglecting for the time being the wear of the pads, is the following, described with reference to FIGS. **2-5**, wherein for the sake of simplicity some elements of the braking system **1** are omitted. For the sake of brevity, the reference numbers are not indicated in FIGS. **3-5**, and not even in FIGS. **6-15**.

(66) In a rest or non-braking condition (FIG. **2**), in the control device **2** and in particular in the master cylinder assembly **7**, the piston **9** is completely rearward in the cylinder **8** and its head **13**—which as said may be provided with the annular sealing gasket **11** (cfr. FIG. **1**) against the wall of the cylinder **8**—, is positioned between the lubrication passage **21***a* and the main communication passage **21** between the cylinder **8** and the tank **20**. The return spring **10** is only slightly biased to ensure the complete return of the piston **9**, the tank **20** is almost full. The brake fluid **5** in the tank **20** is at a high level, at the maximum allowed under the above assumption, and the membrane **25** is in close proximity to the ceiling **22** of the tank **20**. The brake fluid **5** also forms a veil (exaggerated in the Figures) about the piston **9** in order to provide a suitable lubrication.
(67) In the braking device **3** or in the brake caliper, the piston **33** is completely rearward and the pad **31** is out of engagement with the disc W. The gap **35** is completely open, at the maximum allowed extent in the above-mentioned condition of undamaged pads **31**. The gasket **34** (exaggeratedly shown in FIGS. **2-15** for the sake of clarity) is not stressed.
(68) When in the control device **2** the piston **9** starts being thrusted in the compression direction, and as long as its head **13** remains positioned between the lubrication passage **21***a* and the main passage **21**, the compression spring **10** is partially compressed, the brake fluid **5** is forced in the tank **20** through the main passage **21**, and the fluid level in the tank **20** rises until essentially completely filling it.
(69) When, as represented in FIG. **3**, in the control device **2** the head **13** of the piston **9** arrives at the main passage **21** thus closing it, the fluid level in the tank **20** is maximum; the membrane **25** is shown at the “ceiling” **22** of the tank **20**, the filled chamber **26** has a maximum volume substantially corresponding to that of the tank **20**, and the empty chamber **27** has a substantially null minimum volume. In practice, if held at its peripheral region, the membrane **25** (or at least a central region thereof) has a configuration of maximum curvature, with the concavity facing the main passage **21**. In the braking device **3** no changes occur.
(70) In the braking device **3** no changes have occurred yet.
(71) It has to be emphasized that the temporary displacement of the level of the brake fluid **5** in the tank **20** and of the membrane **25** has been intentionally exaggerated in FIGS. **2** and **3** in order to make it evident, but in practice the level increase of liquid **5** during the initial phase of the stroke of the piston **9**, before its head **13** arrives at the main passage **21**, may also be very small with respect to the total volume of the tank **20**.
(72) As the compression stroke of the piston **9** against the force of the return spring **10** continues, because the main passage **21** is closed, the fluid level in the tank **20** remains unchanged (neglecting possible level adjustment due to the lubrication passage **21***a*) at its raised level, and the brake fluid **5** is thrusted into the conduit **4**, further filling the or each cylinder **32** of the slave cylinder assembly **30**, and thrusting the piston **33** in the braking device **3** toward the disc W; the gasket **34** starts deforming. FIG. **4** represents an arbitrary position of the piston **33** in this condition.
(73) In the condition of maximum actuation of the manual actuation member **6**, shown in FIG. **5**, the fluid level in the tank **20** is still at its raised level, equal to the maximum allowed under the above assumption, and the return spring **10** is relatively compressed. In the braking device **3**, the piston **33** is extended by such a total stroke as to bring the pad **31** in contact with and thrusting against the disc W, thus causing the braking of the bicycle wheel. The elastic deformation of the gasket **34** from its rest state is maximum, corresponding to a predetermined stroke of the piston **33**.
(74) With reference to the same FIGS. **2-5** in the reverse order, upon release of the manual actuation member **6**, in the braking device **3**, the gasket **34** progressively recovers the elastic deformation suffered (recovery also called “roll back”), returning to its initial condition, what corresponds to said predetermined stroke of the piston **33**. The cylinder **32** is progressively emptied and the piston **33** and the pad **31** are returned to the rearward position, of non-contact with the disc W, thus setting the rotation of the bicycle wheel free.
(75) The return action of the return spring **10** in the control device **2** entails the return of the brake

fluid **5** from the braking device **3** back in the cylinder **8** of the master cylinder assembly **7** and, after the head **13** of the piston **9** reaches the main passage **21**, the return of the brake fluid **5** from the tank **20** to the cylinder **8**, with a small emptying of the tank **20** itself. The membrane **25** returns to the initial position and the pressures between tank **20** and cylinder **32** of the braking device **3** are re-balanced.

(76) The lubrication passage **21***a* allows passage of a small amount of brake fluid **5** between the cylinder **8** of the master cylinder assembly **7** and the tank **20** in order to lubricate the piston **9** and the inner wall of the cylinder **8**.

(77) The approach of the membrane **25** to the ceiling **22** is thus temporary and occurs during (at least) a length of the stroke of the piston **9** compressing spring **10** and/or of its return stroke. In particular, the temporary approach starts and ends at the crossing of the main passage **21** by the head **13** of the piston **9** in the cylinder **8** of the master cylinder assembly **7**.

(78) As the pads **31** wear out, namely as the friction material (lining or "compound") wears out due to the friction generated during braking, still assuming that the braking system **1** is undamaged and the amount of brake fluid **5** is correct, at the maximum allowed under the above assumption, the operation of the braking system **1** is the same, however an increasing compression stroke of the piston **33** of the slave cylinder assembly **30** is necessary for the pad or friction element **31** to arrive in contact with and thrusting on the disc W. The gasket **34** indeed allows, in its deformed condition, sliding of the piston **33** in the compression direction toward the disc W, while it does not allow sliding of the piston **33** in the opposed direction. In other words, as the pads **31** wear out, during the compression stroke of the piston **33** beyond the elastic deformation of the gasket **34**, also sliding of the piston **33** occurs, while during the decompression stroke, only recovery of the elastic deformation of the gasket **34** ("roll-back") occurs. In the cylinder **32** of the slave cylinder assembly **30** therefore an increasing amount of brake fluid **5** has to be pumped as the pads **31** wear out, and, the stroke of the piston **9** of the main assembly **7** being equal, the tank **20** empties little by little.

(79) FIGS. **6-9** schematically show the positions corresponding to those of FIGS. **2-5**, for heavily worn out pads. It may be noted that the stroke of the piston **33** during a braking has the same extent both with new pads **31** and with worn out pads **31**, and is dictated by the maximum elastic deformation of the gasket **34** and by its recovery or "roll back". However, when the pads **31** are worn out, the piston **33** protrudes more (also in the rest position) from the cylinder **32** with respect to when the pads **31** are new, by an extent corresponding to the wear of the respective pad **31**. In the cylinder **32** of the slave cylinder assembly **30** there is a correspondingly larger amount of brake fluid **5**, and in the tank **20** of the master cylinder assembly **7** there is a comparatively smaller amount of brake fluid **5**.

(80) With reference to FIG. **6**, which represents the rest condition or condition of no braking, it is seen that the membrane **25** is in close proximity to the bottom **23** of the tank **20**, the filled chamber **26** has a comparatively small volume, that becomes minimum and substantially null when the pads **31** are completely worn out, and the empty chamber **27** has a comparatively large volume, that becomes maximum and substantially equal to that of the tank **20** when the pads **31** are completely worn out. In practice, when the pads **31** are completely worn out, the membrane **25**—if held at its peripheral region—has a configuration of maximum curvature in the direction opposed to that mentioned above with reference to FIG. **3**, namely with the convexity facing toward the main passage **21**. The membrane **25** is at a larger distance from the ceiling **22** of the tank **20** with respect to when the pads **31** are not worn out.

(81) The sliding behavior of the piston **33** of the slave cylinder assembly **30** when the gasket **34** is in condition of maximum deformation, and of the consequent emptying of the tank **20**, may be better understood with reference to FIGS. **10-15** wherein the phenomenon is even more manifest, in that they represent the first braking after a sudden detachment of the pad **31** (or of its compound), what may be considered tantamount to a sudden, heavy wear thereof. FIGS. **10-13** schematically show the positions corresponding to those of FIGS. **2-5**, for new pads; in FIG. **13** in particular the

gasket **34** is at maximum deformation. However, because there is no contact with the disc W, the piston **33** of the slave cylinder assembly **30** continues to advance, and the piston **9** of the master cylinder assembly **7** in the control device **2** continues to advance, further compressing the spring **10**, until when, in the condition shown in FIG. **14**, the piston **33** of the slave cylinder assembly **30** enters into contact with the disc W, and the spring **10** is in the condition of maximum compression. When the manual actuation member **6** is released, the piston **33** moves backward only during the above-mentioned roll-back, the gasket **34** straightens up, the spring thrusts the piston **9** of the master cylinder assembly **7** to end of stroke. An additional amount of brake fluid **5** is drawn from the tank **20**, equal to the volume of the slidingly displacement of the piston **33** of the slave cylinder assembly **30**. Thus, the level in the tank **20** does not return to the initial level of FIG. **10**, rather it lowers further, as shown in FIG. **15** which represents the final condition after the braking. The distance of the membrane **25** from the ceiling is great, similar to the case of heavily worn out pad **31**. The overall displacement of the membrane between the position of FIG. **11** and the position of FIG. **15** is large and comparatively quick with respect to the case of normal wear of the pads **31**.
(82) Turning back to FIG. **1**, the control device **2** of the braking system **1** has a membrane detector device **40**, configured to detect the presence of the membrane **25** in at least one predetermined position in the tank **20** and/or the distance of the membrane **25** from the ceiling **22** of the tank **20**. The device **40** may therefore, in some embodiments thereof, also be called meter device **40** of the distance of the membrane **25** from the ceiling **22** of the tank **20**.
(83) Based on the above observations and as better explained hereinbelow, the Applicant believes that not only the position of the membrane **25** inside of the tank **20**, and in particular its distance from the ceiling **22**, are indicative of the volume of the filled chamber **26** (or of the relative volume between filled chamber **26** and empty chamber **27**) and therefore of the amount of brake fluid **5** contained in the tank **20** as well as of the presence and wear of the pads **31** and in general of the integrity of the hydraulic plant **1** (no leakage of brake fluid **5**), but also that its displacements over time are an indication of the correct assembly thereof, in particular of the correct assembly and therefore of the correct operativeness of the master cylinder assembly **7**, in particular of the fact that the main passage **21** is not clogged and the gasket **11** of the head **13** of the piston **9**, if present, is undamaged and correctly positioned.
(84) In the latter respect, the Applicant has observed that a clogging of the main passage **21** entails the lack of temporary raising of the level of the brake fluid **5** in the tank **20** (cf. FIG. **2-3** or **6-7**, for example) and the lack of temporary approach of the membrane **25** to the ceiling **22** when the head **13** of the piston **9** is at the main passage **21**. The temporary displacement of the membrane **25** is therefore null or in any case less than a predetermined threshold.
(85) If, on the contrary, the gasket **11** is not undamaged or is not correctly positioned, or in general if there is not a sufficient sealing between the head **13** of the piston **9** and the cylinder **8**, then at the main passage **21** a leakage of brake fluid **5** may occur. Therefore, the thrust of the piston **9** through the manual actuation member **6** entails the thrust of further brake fluid **5** into the tank **20**. The level of brake fluid **5** in the tank **20** therefore raises more than in conditions of correct operativeness of the master cylinder assembly **7**. The displacement of the membrane **25**, in particular its approach to the ceiling **22** of the tank **20**, is therefore larger than a predetermined threshold.
(86) Overall, in the case of a correct operativeness of the master cylinder assembly **7**, the temporary approach of the membrane **25** to the ceiling **22** of the tank remains within a predetermined range of approaches; differently in the case of incorrect operativeness.
(87) The membrane detector device **40** may thus embody a transducer device of the volume of the filled chamber **26** and/or a transducer device of the volume of brake fluid **5** present in the hydraulic braking system **1**.
(88) Through the membrane detector device **40**, an evaluation device of the wear and/or of the presence of a friction element/s or pad **31** of the hydraulic braking system **1**, and/or an evaluation device of the correct sealing of hydraulic fluid of the hydraulic braking system **1**, and/or an

evaluation device of the correct operativeness of the master cylinder assembly **7** may be embodied.
(89) The membrane detector device **40** may comprise one or more sensors **41** arranged at the ceiling **22** of the tank **20** and/or one or more sensors **42** arranged at the lateral wall **24** of the tank **20**, if present, as well as possibly one or more detectable members **45** borne by the membrane **25**.
(90) The detectable members **45**, if present, are configured to interact with one or more of the sensors **41**, **42** of the membrane detector device **40**.
(91) When for example the or each sensor **41**, **42** is—or embodies with the detectable element(s) **45** borne by the membrane—a proximity sensor, it detects the presence or respectively the absence of the membrane **25** in its "field of view", and therefore in a predetermined position in the tank **20**.
(92) It is possible to define membrane threshold position **25**, corresponding to a predetermined level, for example a minimum tolerable level, of brake fluid **5** inside the tank **20**, and to provide a single proximity sensor which field of view **46** comprises said membrane threshold position **25**, or a region thereof. During emptying of the tank **20** for example because of the wear of the pads **31**, of a leak of the circuit of the brake fluid **5** or for the fall of a pad **31**, the membrane **25** crosses said threshold position triggering the proximity sensor **41**, **42**, **43**.
(93) By providing for example a plurality of proximity sensors having different fields of view (or, in general, of sensors of presence of the membrane in respective predetermined positions), depicted merely by way of an example by the sensors **41***a*, **41***b*; **42***c*, **42***d*, with the respective fields of view **46***a*, **46***b*, **46***c*, **46***d* in FIG. **16**, it is possible to set different threshold positions, and thus to detect different threshold amounts of brake fluid **5** in the tank **20**.
(94) It is also possible, if the number of proximity sensors is sufficient and if the respective fields of view are substantially contiguous and/or overlapping, to embody a more or less precise distance meter **40** and thus obtain a quantitative indication of the amount of brake fluid **5** present in the tank **20**, according to how many and/or which proximity sensor(s) detect(s) the membrane **25** at any given time.
(95) Alternatively, the or each sensor **41**, **42** may be itself—or embody with the detectable element(s) **45** borne by the membrane—a distance meter, configured to measure the distance between the ceiling **22** of the tank **20** and the membrane **25**.
(96) In order to detect the presence in a predetermined position (threshold position) or to measure the distance, it is preferred for the or each proximity sensor **41** to be arranged, preferably on the ceiling **22** of the tank **20** (sensor **41**), facing an as central as possible region of the membrane **25**, namely as far as possible from its peripheral region fixed to the tank **20**, because in such a central region there is the maximum relative displacement of the membrane **25** as the volume of the filled chamber **26**, respectively of the empty chamber **27**, changes. However, also a placement on the lateral wall **24** (sensor **42**) may be effective.
(97) From a constructive point of view, a suitable proximity sensor **41** may be a magnetic sensor, cooperating with a magnet as a detectable member **45** fixed to and mobile with the membrane **25**. A suitable magnetic sensor is a Reed sensor. In use, until the magnet **45** is close to the magnetic Reed sensor, the latter is for example an open switch; when the magnet **45** is far enough from the magnetic Reed sensor, the latter is no longer affected by the magnetic field and switches state, for example closes. The field of view of the proximity sensor **41** is therefore typically a range of positions including a null distance from the Reed sensor (cf. the fields of view **46***e*, **46***f* associated with the sensors **41***e*, **41***f* exemplified in FIG. **17**). However, the field of view of the Reed sensor, and therefore the minimum distance below which the magnet is detected, depends on its sensitivity. By providing, on the ceiling **22** of the tank **20**, a plurality of Reed sensors **41** having different sensitivity, it is thus possible to implement different threshold positions of the membrane **25**: as long as the magnet **45** is detected by all of the sensors **41**, the distance of the membrane **25** is considered to be above the safety threshold, indicative of a sufficiently full tank **20**; when the Reed sensor having the lowest sensitivity does not detect the magnet **45** anymore, the distance of the membrane **25** is considered to be below a first threshold (tank **20** in condition of "reserve"); and

when not even the Reed sensor having the greatest sensitivity detects the magnet **41** anymore, the distance of the membrane **25** is considered to be below the minimum safety threshold (too empty tank **20**). If the sensors are more than two, intermediate positions of the membrane **25** may be detected. The operation might be the reverse in case of Reed sensors that are open or closed in a dual manner with respect to what has been described.
(98) Adjustment of the sensitivity of a Reed sensor may also be provided for, by mounting it through a precision screw support, as better described hereinbelow with reference to FIG. **22**. In this manner, the final user or an installer of the bicycle braking system **1** is allowed to choose the or each level threshold, corresponding for example to a desired wear degree of the pads.
(99) In case of leakage of brake fluid or of a fall of the pads **31**, the level in the tank may quickly change, the membrane **25** being for example first detected by all of the Reed sensors, and immediately thereafter only by the one having the greatest sensitivity.
(100) Alternatively to a Reed sensor, another type of magnetic sensor may also be used, capable of detecting a movement or a position of a magnet. For example, a Hall sensor may be used, in particular a 3D Hall sensor, or a magneto resistive sensor, such as for example an AMR ("Anisotropic magneto resistive") sensor, a GMR ("Giant magneto resistive") sensor or a TMR ("Tunnel magneto resistive") sensor. All the above-mentioned sensors detect magnetic fields and output signals relating to the position, angle, force and/or direction of the detected magnetic field, thus also allowing the meter device **40** to be configured; in the case of the 3D sensors, the movement in space of a magnetic field may be detected. Furthermore, advantageously these sensors are capable of performing high-precision measurements of the magnetic field still having an extremely compact footprint and low energy consumption.
(101) Instead of a magnet **45**, magnetized rubber may be used for the membrane **25**.
(102) With reference to FIGS. **18-19**, preferably in the case of use of a magnetic sensor, in the membrane **25**—which shape is merely by way of an example—a pocket **47** is formed, comprising two overlapping layers and a cut **48**, for example straight or circular, in one of the layers, preferably in that on the side of the empty chamber **27** of the tank **20**. The pocket **47** may be made for example by providing a "mushroom-like" protrusion in a mold of the membrane **25**.
(103) The magnet **45** is forced in the pocket **47** through the cut **48**, and, thanks to the elasticity of the membrane **25**, the latter closes onto the magnet **45**, keeping it in the pocket **47**. According to the mutual thickness of the overlapping layers, it is possible to make the magnet **45** protrude more or less on either side of the membrane **25** and thus to cause its volume to be detracted from the volume of the filled chamber **26** (as in the exemplifying case of FIG. **19**) or from the volume of the empty chamber **27**. Alternatively, the magnet **45** may be glued or otherwise fixed to the membrane **25**, on either side.
(104) A suitable proximity sensor **41 42** may also be an optical sensor. An optical sensor comprises in general a light source, preferably an LED, and a photodiode or phototransistor located to receive the light emitted by the light source and reflected by the membrane **25** (or by a detectable member **45** borne by the membrane) when it is in the field of view of the optical sensor.
(105) The characteristics of the optical sensor and/or of a polarizing circuit thereof may be chosen so that the intensity of the light reflected by the membrane **25** and received by the photodiode or phototransistor, and thus the response of the optical sensor, changes as the position of the membrane **25** within the field of view, which may have an extent even comparable to a maximum size of the tank **20**, Changes.
(106) Also a single optical sensor may thus configure a distance meter **40**.
(107) In case of use of an optical sensor, it may be appropriate to provide for a reflective insert, or in any case made of more reflecting material than the material forming the membrane **25**, as a detectable member **45** on the membrane **25**. Such a (comparatively) reflective insert may be glued on the membrane **25** or be inserted in a pocket **47** analogously to what has been disclosed with reference to the magnet.

(108) Alternatively, a colored membrane may be used. In the cases of an infrared optical sensor, a red membrane has turned out to be preferable.
(109) It should be noted that the response of the optical sensor may also depend on other characteristics of the membrane **25**, such as the material, the color, the local curvature, but all these factors may be suitably taken into account, possibly providing for a suitable processing of the output signal of the optical sensor.
(110) The optical sensor may be located on the inner surface of the ceiling **22** of the tank **20**, or on its outer surface in case the ceiling **22** is transparent.
(111) A plurality of sensors of a different type may be provided for. For example, a distance meter having a field of view corresponding to a large distance range from the ceiling **22** of the tank and a proximity sensor having a field of view corresponding to a short distance range may be provided for, or vice versa.
(112) Summing up, the membrane detector device **40** may provide a two-level output, corresponding to an amount of brake fluid **5** in the tank **20** larger or smaller than a predetermined threshold, an output having a number of discrete levels, corresponding to an amount of brake fluid **5** smaller than a given number of predetermined thresholds, or an output, having several discrete levels or analogue, indicative of the amount of brake fluid **5** present in the tank **20**.
(113) Furthermore, the membrane detector device **40** may provide an instantaneous output and/or an output processed based on two or more detections over time.
(114) As explained above, if the amount of brake fluid **5** in the braking system **1** is not sufficient, for example because the hydraulic braking system **1** is not undamaged, then the level of brake fluid **5** in the tank **20** never rises to the maximum foreseen level—corresponding as explained above to the instant when the head **13** of the piston **9** is at the main passage **21**—, namely the filled chamber **26** is never at the maximum foreseen volume, or the membrane **25** is never at the minimum, substantially null distance from the ceiling **22** of the tank **20**. In case of leakage of brake fluid **5**, the volume of the filled chamber **26** decreases—more or less quickly according to the extent of the leakage—even down to zero, and the membrane **25** reaches the bottom **23** of the tank **20**, namely the maximum distance from the ceiling **22** of the tank **20**. In case of leakage at the head **13** of the piston **9**, for example in case of damage or incorrect positioning of the gasket **11**, the membrane **25** moves closer the ceiling **22** than in case of correct operativeness of the master cylinder assembly **7**. In case of clogging of the main passage **21**, the membrane **25** does not temporarily move close to the ceiling **22** during the length of the stroke of the piston **9** after the head **13** of the piston **8** passes by the main passage **21**.
(115) The output signal of the membrane detector device **40** and/or its change over time and/or its rate of change may advantageously be used for the diagnosis of the braking system **1** and in particular of its master cylinder assembly **7**, upon leaving the factory (as quality control) or after replacement of the pads **31** or after a bleeding operation, possibly temporarily replacing the pads **31** with a spacer of the pistons **33** of the braking device **3**, as well as during the use of the hydraulic system **1** in order to check for intervened leaks and/or the wear of the pads **31** and/or their accidental fall and/or the other possible drawbacks mentioned above.
(116) The membrane detector device **40** may also comprise or be associated with memory means **50**, embodied by discrete components, for one or more historical detected values.
(117) The membrane detector device **40** may also comprise or be associated with a data processing system **51** for processing its instantaneous and/or historical output.
(118) The data processing system may comprise electric components and/or discrete electronic components and/or a micro-controller, which may also embody the memory means **50**.
(119) The processing may comprise, for example, the evaluation of a current detected value and/or the comparison of at least one historical detected value with at least one current detected value and/or the computation of a rate of change of the detected value.
(120) The data processing system may be configured to: evaluate a volume of the filled chamber **26**

based on said at least one output of the membrane detector device **40** and output at least one signal when the evaluated volume of the filled chamber **26** is less than at least one volume threshold, and/or evaluate a displacement speed of the membrane **25** based on said at least one output of the membrane detector device **40** in at least two successive time instants and issue a signal when the displacement speed is higher than a maximum speed threshold and/or evaluate the position of the membrane **25** in the tank **20** based on the output of the membrane detector device **40** and/or evaluate a displacement amplitude of the membrane based on said at least one output of the membrane detector device in at least two successive time instants and/or evaluate the wear and/or the presence of at least one friction element **31** of the hydraulic braking device **3** based on said at least one output of the membrane detector device **40** and/or evaluate the correct sealing of the hydraulic fluid **5** in the brake fluid circuit of the hydraulic braking system **1** based on said at least one output of the membrane detector device **40**, for example based on the position of the membrane **25** in the tank **20** and/or of its displacement speed over time.
(121) Evaluating the wear may comprise for example establishing that the wear is higher than a predetermined wear threshold when the distance of the membrane **25** from the ceiling **22** of the tank **20** is larger than a predetermined maximum distance threshold and/or may further comprise establishing that the wear is greater than a predetermined alert wear threshold when the distance of the membrane **25** from the ceiling **22** of the tank **20** is less than the predetermined maximum distance threshold and greater than a predetermined alert distance threshold.
(122) Evaluating the correct sealing of the hydraulic fluid may comprise for example verifying that the displacement speed over time of the membrane **25** from the ceiling **22** of the tank **20** is less than a predetermined speed threshold.
(123) Alternatively or additionally, the data processing system **51** may be configured to: evaluate whether an output signal of the meter device **40** of the distance of the membrane **25** from the ceiling **22** of the tank **20** is indicative of a temporary approach of the membrane **25** to the ceiling **22**, in particular having an extent corresponding to a predetermined approach range, and determine that the master cylinder assembly **7** is correctly operating if the evaluation is positive, that the master cylinder assembly **7** is not correctly operating if the evaluation is negative.
(124) In this manner the data processing system **51** embodies a diagnosis of the bicycle hydraulic braking system **1** of the disc type, in particular of its master cylinder assembly **7**, wherein under term “diagnosis”, the examination aimed at formulating a judgment on the conditions and the operation of the various parts is meant to be indicated, including testing.
(125) The data processing system **51** may be part of the control device **2** as shown, or may be part of a different component of the hydraulic braking system **1**.
(126) The control device **2** may also be part of a bicycle equipment comprising, besides the hydraulic braking system, also a speed gear shifting system (not shown). In that case, the control device **2** may also comprise at least one second manual actuation member configured to issue a gear ratio change command (cf. for example the gearshift levers **70**, **71** in FIG. **23** subsequently described). In that case, the data processing system **51** may be part of a component of the speed gear shifting system, for example may be part of a gearshift assembly thereof, associated with the hub of the rear wheel.
(127) The membrane detector device **40** may also comprise or be associated with one or more output devices **52**.
(128) For example, the data processing system **51** may be configured to issue an alarm signal when the wear of the friction element **31** is higher than a maximum wear threshold and/or the at least one friction element is absent and/or the sealing of the hydraulic fluid **5** is not correct and/or the master cylinder assembly **7** is not correctly operative.
(129) Merely by way of an example, in FIG. **1** there are shown, in a totally schematic manner, a luminous indicator **53**, an audible indicator **54**, and a communication device **56** in communication with a second communication device (not shown), none of which is strictly necessary.

(130) The membrane detector device **40** may also comprise or be associated with a power supply source **57** which provides for power supplying the above components.
(131) Also the output devices **52** or some of them may be part of the control device **2** as shown, or part of a different component of the hydraulic braking system **1**, or part of a component of the speed gear shifting system, for example of the gearshift assembly associated with the hub of the rear wheel.
(132) In that case, the output device **52** may use a luminous and/or audible indicator also intended for issuing signals relating to the gearshift or other bicycle equipment, for example with a suitable control by the micro-controller **51**, and/or the power supply source **57** may be intended also for powering other components of the control device **1**.
(133) The communication device **56** may be for example a communication module, preferably a short range and low consumption one, for example according to the Bluetooth, Bluetooth Low Energy, or ANT+ protocol.
(134) The luminous indicator **53** may comprise one or more LEDs or other luminous sources. A single luminous indicator may for example light up as soon as the level of brake fluid **5** in the tank **20** lowers below a predetermined threshold level. When there is a polychromatic LED or another polychromatic source, or there are plural LEDs or other sources of different color, or there is an LED array, it is possible to provide a multiple visual indication, for example by turning on an orange light or a flashing light when the level of brake fluid **5** lowers below a first threshold and a red light or a steady light when the level of fluid **5** lowers below a second threshold less than the first threshold; or to provide a visual indication of the amount of brake fluid **5** contained in the tank **20**, by turning off the LEDs at the top or right end of the array of LEDs as the tank **20** empties.
(135) The audible indicator **54** may be for example a small buzzer and may emit one or more sounds for providing a more or less accurate communication relative to the level of the brake fluid **5** in the tank **20**.
(136) The battery power supply source **57** may be dedicated to the membrane detector device **40** or may be intended also for powering other electric or electronic devices housed in the control device **2**, possibly including the above-mentioned data processing system **51**, memory means **50** and/or output devices **52**. Alternatively, the battery power supply source **57** may advantageously be that of a cycle computer or of a smartphone, a tablet or other computer connected in a fixed or removable manner to the control device **2**. Merely by way of an example, the communication may occur through a micro-USB connector.
(137) In order to contain the energy consumption as much as possible, the membrane detector device **40** may be activated only periodically, under the control of said data processing system **51**, for example at predetermined time intervals or implementing a braking counter, for example counting pulses generated by a micro-switch activated upon pressure of the manual actuation member **6**, once a day or once a week or once a month, etc.
(138) Alternatively or additionally, the membrane detector device **40** or in general the control device **2** may have a stand-by mode and the tank **20** or in general the control device **2** may be provided with a waking system, for example based on an accelerometer.
(139) It may also be provided for the membrane detector device **40** to be selectively actuatable by the cyclist.
(140) The control device **2** may also comprise, alternatively or additionally to the battery power supply source **57**, an “energy harvesting” system, which exploits for example solar energy, kinetic energy or something else.
(141) Advantageously, one or more of the sensors **41**, **42** and one or more of the output devices **52** are housed on opposite faces of a single printed circuit board or PCB **60**, as shown by way of an example in FIGS. **20-21**. Powering connectors **61** leading to the power supply source **55** are also shown.
(142) FIG. **22** shows, by way of an example, a PCB **60** carrying a precision screw support **62** for

mounting a sensor of the membrane detector device **40** in order to accomplish the sensitivity adjustment.
(143) FIGS. **23-27** show, by way of an example, a practical embodiment of the main components of a braking system **1** as described above. Analogous component are indicated with the same reference number.
(144) As far as the control device **2** is concerned, in FIGS. **23-24** a body **130** may be recognized, configured for fixing to the bicycle handlebar, possibly grippable, wherein the master cylinder assembly **7** is formed or fixed, and on which the manual actuation member **6** or brake lever is articulated. Also the above mentioned gearshift levers **70**, **71** are visible, as an example of manual actuation members of a speed gear shifting system.
(145) The PCB **60** is preferably fixed, as shown by way of an example in FIGS. **23-24**, in a suitable position which is adjacent to the ceiling **22** or to the lateral wall **24** of the tank **20** and adjacent to the outer surface of the control device **2**.
(146) If the control device **2** is covered by a protection sheath (not shown), the latter may be provided with a suitable aperture at the or each output device **52**, and with possible protection windows, for example a transparent window for protecting the one or more luminous indicators **53**.
(147) When a luminous indicator **53** or an audible indicator **54** is provided for, preferably the PCB **60** is adjacent to the outer surface of the control device **2** in a proximal position to the cyclist in the condition of use of the bicycle and in particular visible by the cyclist, for example on the proximal face of a control device for a curved handlebar, of the so-called “drop-bar” type, as shown.
(148) A kinematics **131** may also be seen, interposed between the manual actuation member **6** or brake lever and the cylinder **8** of the master cylinder assembly **7**, more evident in FIG. **25**.
(149) In the master cylinder assembly **7**, the above discussed components may be seen, and additionally a cap **132** configured to hold the gasket **11** (if provided for) on the head **13** of the piston **9** and to support the return spring **10**. The cap **132** is not strictly necessary.
(150) Advantageously, the tank **20** is formed in part by a region **133** one piece with the body **130** and in part by a cover **134**, although this is not strictly necessary. Advantageously, the peripheral region **135** of the membrane **25** is watertight-sealing clamped between the cover **134** and the region **133** one piece with the body **50** of the control device **2**.
(151) It may be seen, also with reference to FIGS. **18-19**, that the example membrane **25** has a substantially tub-like shape, and has a central region **136** having a substantially flat zone **137** intended to conform with the ceiling **22** in the condition of filled tank **20** and respectively with the bottom **23** in the condition of empty tank **20**, said peripheral region **135** intended to be fixed to the tank **20**, and an annular region **138** interposed therebetween, which according to the amount of brake fluid **5** present is intended to conform with and adhere to a variable region of the lateral wall **24** of the tank **20**. The peripheral region **135** and the substantially flat zone **137** of the central region **136** extend in planes that remain substantially parallel, while the annular region **138** extends substantially orthogonally to said planes, in a non-stressed condition of the membrane **25**.
(152) The region **133** one piece with the body **130** defines the bottom **23** of the tank **20** and, preferably, a first part of its lateral wall **24**. The cover **134** defines the ceiling **22** of the tank **20** and, preferably, the remaining part of its lateral wall **24**. The region **133** one piece with the body **130** and the cover **134** are thus both concave, and have the concavities facing towards each other.
(153) Preferably, the region **133** one piece with the body **130** and the cover **134** have approximately the same depth—or height of the respective part of lateral wall **24** of the tank **20**—so that the membrane **25** adopts a curvature substantially the mirror image in its two extreme positions at the ceiling **22** and at the bottom **23** of the tank **20**.
(154) For further details relating to the control device **2** and to the tank **20** reference is made to the above cited document EP3835191A1, as if it were directly incorporated herein.
(155) It is noted that the tank **20** might also have a spheric or almost spheric shape, wherein the membrane **25** is peripherally fixed to a diametric plane of the sphere and adopts a convex

configuration when the tank is filled and a concave configuration when the tank is empty, a lateral wall being absent or substantially absent.
(156) An example braking device **3** or brake caliper is represented in FIGS. **26-27**. The brake caliper **3** is shown upside down for the sake of convenience; in FIG. **26** the pad **31** is omitted, visible instead in FIG. **27**. The brake caliper **3** has a body **141** configured for fixing to the bicycle frame or on the fork in proximity to the hub of a wheel with which the disc W (FIG. **1**) integrally rotates. In the body **141**, the gap **35** in which the above-mentioned disc W is inserted, as well as (FIG. **27**) one of the two pistons **33** of the slave cylinder assembly **30**, housed in the respective cylinder **32**, may be seen.
(157) The various embodiments, variants and/or possibilities of each component or group of components that have been described are to be meant as combinable with each other in any manner, unless they are mutually incompatible.
(158) The above is a description of various embodiments, variants and/or possibilities of inventive aspects, and further changes can be made without departing from the scope of the present invention. The shape and/or size and/or location and/or orientation of the various components and/or the succession of the various steps can be changed. The functions of an element or module can be carried out by two or more components or modules, and vice-versa. Components shown directly connected to or contacting each other can have intermediate structures arranged in between them. Steps shown directly following each other can have intermediate steps carried out between them. The details shown in a figure and/or described with reference to a figure or to an embodiment can apply in other figures or embodiments. Not all of the details shown in a figure or described in a same context must necessarily be present in a same embodiment. Features or aspects that turn out to be innovative with respect to the prior art, alone or in combination with other features, should be deemed to be described per se, irrespective of what is explicitly described as innovative.

Claims

1. A method for diagnosing a bicycle hydraulic disc braking system, the method comprising the steps of: a) providing a master cylinder assembly comprising a cylinder, with a piston that reciprocates inside the cylinder against the action of a return spring, and a hydraulic tank having a bottom, a ceiling, and a membrane that divides the tank into a variable volume filled chamber and a complementarily variable volume empty chamber, said hydraulic tank being fluidically connected to the cylinder through at least one main passage in said bottom, b) providing a metering device for measuring a distance of the membrane from the ceiling of the hydraulic tank, c) inserting a hydraulic fluid in the master cylinder assembly, d) reciprocating the piston inside the cylinder, e) evaluating whether an output signal of the metering device is indicative of a temporary approach of the membrane to the ceiling having an extent corresponding to a predetermined approach range, and, f) establishing that the master cylinder assembly is correctly operating when the output signal is indicative of a temporary approach of the membrane to the ceiling.
2. The method according to claim 1, wherein the master cylinder assembly is fluidically connected in the hydraulic braking system during the execution of steps d), e), f).
3. A diagnostic system for a bicycle hydraulic disc braking system, the diagnostic system comprising: a master cylinder assembly having a cylinder, a piston that reciprocates inside the cylinder against the action of a return spring and a hydraulic tank comprising a bottom, a ceiling, and a membrane that divides the hydraulic tank into a variable volume filled chamber and a complementarily variable volume empty chamber, said hydraulic tank being fluidically connected to the cylinder through at least one main passage in said bottom, a hydraulic fluid filling at least in part the master cylinder assembly, a metering device for outputting a signal based on a measurement of a distance between the membrane and the ceiling of the tank, a data processing system configured to establish that the master cylinder assembly is correctly operating when the

output signal is indicative of a temporary approach of the membrane to the ceiling having an extent corresponding to a predetermined approach range.

4. The system according to claim 3, wherein the temporary approach starts and ends at the crossing of the main passage by a head of the piston in the cylinder of the master cylinder assembly.

5. The system according to claim 3, wherein the metering device comprises a sensor arranged at the ceiling of the tank and a detectable member borne by the membrane.

6. The system according to claim 5, wherein the sensor is an optical sensor, comprising a light source and a photodiode or phototransistor located to receive the light emitted by the light source and reflected by the membrane or by said detectable member borne by the membrane when said membrane or said detectable member is in the field of view of the optical sensor.

7. The system according to claim 6, wherein the detectable member borne by the membrane is an insert that is more reflective than the membrane.

8. The system according to claim 6, wherein the light source is an LED.

9. The system according to claim 5, wherein in the membrane a pocket is formed, configured to house the detectable member borne by the membrane.

10. The system according to claim 3, wherein the metering device is configured to be actuatable and is: selectively actuatable by a user and/or periodically actuated under the control of a controller.

11. The system according to claim 10, wherein the metering device has a stand-by mode and a control device is provided with a waking system.

12. The system according to claim 11, wherein-said at least one output device is selected from the group consisting of: at least one audible indicator, at least one luminous indicator, and a communication device.

13. The system according to claim 12, wherein at least one sensor and said at least one luminous indicator are housed on opposite faces of one and a same printed circuit board.

14. The system according to claim 3, comprising at least one output device outputting the outcome of said evaluation.

15. The system according to claim 3, wherein the piston of the master cylinder assembly has a head and a thrust end and a gasket is positioned at the head.

16. A manual control device for a bicycle hydraulic disc braking system, the control device comprising a manual actuation member configured to issue a braking command and the diagnostic system according to claim 3, wherein the piston is displaced inside the cylinder in a forward stroke in response to the actuation of the manual actuation member.

17. A bicycle hydraulic disc braking system, comprising: the master cylinder assembly and the metering device of the diagnostic system according to claim 3, wherein the piston is displaced inside the cylinder in a forward stroke in response to the actuation of the manual actuation member, at least one control device, the at least one control device comprising a manual actuation member, the manual actuation member configured to issue a braking command, and a respective hydraulic braking device fluidically connected to the master cylinder assembly, wherein said diagnostic system is part of a component of the hydraulic disc braking system.

18. The bicycle hydraulic disc braking system according to claim 17, wherein the data processing system is part of the control device.

19. A bicycle comprising: a hydraulic disc braking system comprising: the diagnostic system according to claim 3, wherein the piston is displaced inside the cylinder in a forward stroke in response to the actuation of the manual actuator, at least one control device, the at least one control device comprising a manual actuation member, the manual actuation member configured to issue a braking command, a respective hydraulic braking device fluidically connected to the master cylinder assembly, and a speed gear shifting system, said at least one control device comprising at least a second manual actuator configured to issue a speed gear shifting command to the speed gear shifting system, and wherein the data processing system is part of a component of the speed gear shifting system.

20. The bicycle according to claim 19, wherein the data processing system is part of a gearshift assembly of the speed gear shifting system associated with a hub of a bicycle rear wheel.